

Semantic Grounding on Memory-Retrieval-Based Video World Models

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Abstract

Memory-retrieval video world models such as WorldPack improve long-horizon spatial consistency by retrieving geometrically relevant past frames into the generation context. We identify two limitations of WorldPack’s purely geometric retrieval and propose targeted improvements: (i) a *direction-toward* retrieval term that recovers frontal-view frames whose viewing direction points at the current generation position, and (ii) *patch-level memory grounding*, which augments the target-memory cross-attention with geometric ray-overlap and frozen DINOv3 semantic similarity so the model can align *which patches* of a retrieved frame correspond to the frame being generated. On the LoopNav benchmark, we show these improvements are *horizon-dependent*: they give no benefit at short navigation ranges, where recent context already covers the loop, but at long ranges (nav=30) the full method improves over the WorldPack baseline on all four metrics, including a 22% reduction in Loop-Closure Error, the measure most directly tied to spatial memory.

1 Introduction

Video world models generate future observations conditioned on past frames and actions, and are increasingly used as learned simulators for navigation and control. A central challenge is *long-horizon spatial consistency*: when an agent revisits a previously observed location, the model should reconstruct it faithfully rather than drift. WorldPack [1] addresses this by (a) *retrieving* spatially relevant past frames using a geometric scoring function and (b) *packing* them into a fixed-length context via multi-resolution compression, on top of the CDiT backbone of NWM [2].

We build on this line of work and make two observations. First, WorldPack’s retrieval score is grounded entirely in the current frame’s *outgoing* viewing direction and does not represent past frames that *look toward* the current position, a complementary frontal source of scene structure. Second, retrieval alone selects *which* frames enter memory but provides no mechanism for the model to determine *which patches* of a retrieved frame correspond to the target, which is difficult when the memory frame observed the location from a different viewpoint. We address the first with an added retrieval term and the second with patch-level memory grounding.

2 Background: WorldPack Retrieval

For a past frame i at position $\mathbf{p}_i$ with unit viewing direction $\mathbf{d}_i$, and current generation position $\mathbf{p}$ with direction $\mathbf{d}$, WorldPack defines forward projection $s_i = (\mathbf{p}_i - \mathbf{p})^\top \mathbf{d}$, lateral offset ℓ_i , and

directional similarity $\cos \Delta\theta_i = \mathbf{d}_i^\top \mathbf{d}$. The score combines a co-visibility term (w_c) and a look-back term (w_a):

$$\begin{aligned} \text{score}_i = & w_c \cdot \max(\cos \Delta\theta_i, 0) \exp\left(\frac{-s_i^2}{2\sigma_s^2}\right) \exp\left(\frac{-\ell_i^2}{2\sigma_\ell^2}\right) \\ & + w_a \cdot \max(-\cos \Delta\theta_i, 0) \exp\left(\frac{-(s_i - \mu_s)^2}{2\sigma_s^2}\right) \exp\left(\frac{-\ell_i^2}{2\sigma_\ell^2}\right). \end{aligned} \quad (1)$$

Both terms depend on the current frame’s outgoing direction $\mathbf{d}$, leaving frames whose direction $\mathbf{d}_i$ points *toward* the current position unrepresented.

3 Method

3.1 Direction-Toward Retrieval Term

Motivation. Past frames whose viewing direction points toward the current generation position offer complementary *frontal-view* information, analogous to seeing someone across a corridor face-on as you approach. Their viewing rays intersect the current scene from the opposite side, providing structure at the generation point that neither w_c nor w_a captures.

Formulation. For past frame i and generation position $\mathbf{p}$ we define, along the past frame’s *own* viewing axis,

$$s'_i = (\mathbf{p}_i - \mathbf{p})^\top \mathbf{d}_i, \quad \ell'_i = \|(\mathbf{p}_i - \mathbf{p}) - s'_i \mathbf{d}_i\|, \quad \cos \Delta\theta'_i = \frac{\mathbf{d}_i^\top (\mathbf{p}_i - \mathbf{p})}{\|\mathbf{p}_i - \mathbf{p}\|}. \quad (2)$$

We add a third term:

$$w_o \cdot \max(\cos \Delta\theta'_i, 0) \exp\left(\frac{-(s'_i - \mu_{s'})^2}{2\sigma_{s'}^2}\right) \exp\left(\frac{-\ell'^2_i}{2\sigma_{\ell'}^2}\right). \quad (3)$$

Key design choice: broad forward Gaussian. Critically, $\sigma_{s'} \gg \sigma_s$ (we use $\sigma_{s'} = 5.0$ vs. $\sigma_s = 0.01$). A frame looking toward the current position can be at a wide range of distances and still be useful; it need not be tightly co-located. A retrieval-only analysis on nav=30 trajectories confirms this: relative to $\sigma_{s'} = 1.0$, the broad Gaussian surfaces ~ 8 additional look-back frames per query at mean forward distance $s'_i \approx 8$ with near-perfect directional alignment ($\cos \Delta\theta'_i \approx 0.96$), precisely the far loop-closure frames a narrow window discards. This distinction proves decisive at long horizons (Section 4).

3.2 Patch-Level Memory Grounding

Retrieval selects *which* frames enter memory; the model must still infer *which patches* of a retrieved frame correspond to the target being generated, which is hard when the memory frame observed the location from a different viewpoint (e.g. on the return leg of a loop). We add two patch-level bias terms to the target–memory cross-attention. Let the target frame have patches indexed by p and the concatenated retrieved-memory keys have patches indexed by q . Standard scaled-dot-product cross-attention forms the logit $z_{pq} = \mathbf{q}_p^\top \mathbf{k}_q / \sqrt{d}$; we augment it as follows.

Geometric ray-overlap bias. For a frame with camera orientation (yaw θ , pitch ϕ), each patch j has a world-space unit viewing ray $\mathbf{r}_j \in \mathbb{S}^2$, obtained by mapping the patch-centre pixel direction through the pinhole intrinsics and rotating it into the world frame by the camera rotation;

equivalently $\mathbf{r}_j = R_j \mathbf{e}_3$, the third column of the per-patch orientation matrix $R_j \in SO(3)$. The geometric bias is the cosine of the angle between the target and memory patch rays:

$$B_{pq}^{\text{geo}} = (\mathbf{r}_p^{\text{tgt}})^\top \mathbf{r}_q^{\text{mem}}, \quad \mathbf{r}_p^{\text{tgt}}, \mathbf{r}_q^{\text{mem}} \in \mathbb{S}^2. \quad (4)$$

Because it depends only on camera pose and not on pixel content, B^{geo} is well-defined for the *not-yet-generated* target frame.

Semantic bias. Let $\mathbf{f}_j = \text{DINOv3}(I)_j / \|\text{DINOv3}(I)_j\|$ be the ℓ_2 -normalised frozen DINOv3 [4] patch embedding for patch j of image I . Since the target is not yet generated, we use the most recent context frame I^{ctx} as its proxy (a position-aligned stand-in under small inter-frame motion, and computable identically at training and rollout time). The semantic bias is the patch cosine similarity:

$$B_{pq}^{\text{sem}} = (\mathbf{f}_p^{\text{ctx}})^\top \mathbf{f}_q^{\text{mem}}. \quad (5)$$

Gated, memory-only application. For attention head h with learnable scalar gates $g_h^{\text{geo}}, g_h^{\text{sem}}$ (both initialised to 0), the biased attention weight is

$$\alpha_{pq}^h = \text{softmax}_q \left(\frac{\mathbf{q}_p^h \mathbf{k}_q^h}{\sqrt{d}} + m_q \left(g_h^{\text{geo}} B_{pq}^{\text{geo}} + g_h^{\text{sem}} B_{pq}^{\text{sem}} \right) \right), \quad (6)$$

where $m_q \in \{0, 1\}$ is the *memory-only* mask: $m_q = 1$ for keys belonging to retrieved memory frames and $m_q = 0$ for recent-context keys. Zero-initialised gates make the model identical to the base at the start of training, so no warm-up is required and the biases are learned only if they help.

The memory-only mask is essential. During autoregressive rollout the recent-context frames are the model’s *own drifting generations*; biasing attention toward them ($m_q = 1$ on context) amplifies error, whereas restricting the bias to the clean, pre-horizon retrieved memory re-anchors the target and reduces drift.

Diagnostic: drift, not retrieval, is the bottleneck. A teacher-forced probe, feeding ground-truth context frames instead of generated ones, shows single-step prediction is strong (SSIM ≈ 0.52) while free-running rollout collapses (SSIM ≈ 0.32), identifying *autoregressive drift* as the dominant error source. Memory grounding targets this directly, motivating the memory-only design above.

4 Experiments

Setup. We evaluate on the LoopNav benchmark [3] (Minecraft loop navigation), ABA task: the agent explores $A \rightarrow B$ (context) and must reconstruct the return $B \rightarrow A$ (evaluated). The *navigation range* controls loop length. We report SSIM and PSNR (distortion), LPIPS [5] (perceptual), and Loop-Closure Error (LCE), our task metric: the LPIPS between the true starting frame and the generated frame at the returned pose, which directly measures whether memory brought the model back to the correct location. All models fine-tune the CDiT-XL backbone.

Retrieval term, short horizon. At nav=5, 20k steps, without architectural changes, the added term improves SSIM and LPIPS at a small PSNR cost, though margins are within noise at this horizon (Table 1).

ViewRoPE and trajectory packing. We explored two architectural extensions: ViewRoPE (SO(3) rotation of cross-attention Q/K by camera-ray direction) and trajectory packing (multi-resolution memory compression). Neither exceeded the baseline at nav=5: ViewRoPE degraded

Table 1: 2-term vs. 3-term scorer, 20k steps, ABA, nav=5.

Metric	2-term	3-term	Winner
SSIM $\uparrow$	0.3506	0.3628	3-term
PSNR $\uparrow$	12.48	12.28	2-term
LPIPS $\downarrow$	0.7089	0.6866	3-term

when applied to a converged checkpoint and only recovered after ~ 100 k steps of joint training, and packing matched but did not surpass the baseline. We attribute this to the *navigation horizon*: at nav=5 short loops are already covered by recent context, so no memory mechanism can demonstrate benefit, consistent with WorldPack’s own results where gains appear only at larger ranges. This motivates evaluation at longer horizons.

Long horizon: patch-level grounding wins. At nav=30 the return path is far from the exploration phase and drift dominates. Models are trained 40k steps and evaluated with the corrected $\sigma_{s'} = 5.0$ scorer (Table 2).

Table 2: ABA, nav=30, 40k steps ($n=15$ eval trajectories). Grounding = geometric + semantic patch-level memory grounding (memory-only). Full method = 3-term retrieval + grounding.

Config	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	LCE $\downarrow$
2-term (baseline)	0.2313	9.72	0.9442	1.0044
3-term (retrieval)	0.2313	10.32	0.8774	0.9083
2-term + grounding	0.3300	11.18	0.7918	0.8276
3-term + grounding (full)	0.3612	11.78	0.8258	0.8438

The full method improves over the WorldPack baseline on *all four* metrics: SSIM $0.23 \rightarrow 0.36$, PSNR $9.72 \rightarrow 11.78$, LPIPS $0.94 \rightarrow 0.83$, and LCE $1.00 \rightarrow 0.84$ ($\sim 22\%$ relative). Ablation shows patch-level grounding is the primary driver: it alone (2-term + grounding) already beats the baseline on every metric, while the retrieval term adds a further SSIM/PSNR gain. Together with the null result at nav=5, this establishes a *horizon-dependent* benefit: grounding matters precisely when loops are long enough that recent context cannot cover the return and drift dominates, with the largest relative gain on Loop-Closure Error.

On the corrected retrieval σ . An earlier configuration ran the w_o term with a narrow $\sigma_{s'} = 1.0$ rather than the intended 5.0. The narrow window rejected the far look-back frames the term is designed to capture and *degraded* performance below the baseline (SSIM 0.2011 vs. 0.2313). Correcting to $\sigma_{s'} = 5.0$ removes this penalty, underscoring that the retrieval window must be broad enough to reach distant loop-closure frames.

5 Discussion and Ongoing Work

The results support a simple picture: at long horizons the bottleneck is autoregressive drift, and re-anchoring the generation to clean, correctly-retrieved memory, at the patch level, directly reduces it, most visibly in loop closure. The retrieval term and the grounding are complementary: retrieval must surface the right frames (which the broad- σ correction enables), and grounding makes those frames usable despite viewpoint change.

Results above use 40k training steps and $n=15$ evaluation trajectories. A 100k-step run and an expanded $n=40$ evaluation are in progress to tighten statistical significance; preliminary trends are consistent with the 40k results. Further work includes extending the evaluation to larger navigation ranges (nav=50) to trace the full horizon-dependence curve, and to the ABCA spatial-reasoning task.

References

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